

Inertial Relativity Mathematics

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Category	Relation
Gravitational Time Dilation	
	$\frac{d\tau}{dt} = \sqrt{1 - k}$
Raw Time Dilation	
	$\text{RTD} = \sqrt{k}$ $\text{RTD}^2 = k$
Space-Time Equivalence (STE)	
	$\frac{\text{DTD}_1}{\text{DTD}_2} = \sqrt{\frac{I_1}{I_2}} \cdot \left(\frac{R_2}{R_1}\right)^3$
Isometric Equivalence (constant density)	
	$k = \left(\frac{I_2}{I_1}\right)^{\frac{1}{5}}$ $k = \left(\frac{\frac{2}{5}m_2r_2^2}{\frac{2}{5}m_1r_1^2}\right)^{\frac{1}{5}}$
Static Density (Isometric)	
	$k = \left(\frac{M_2}{M_1}\right)^{\frac{1}{3}}$ $M' = k^3 M$ $R' = kR$
Static Mass (Gravitational)	
	$k = \frac{R_S}{R}$ $R' = kR$ $R_S = kR$
Inertial Density	
	$k = \frac{I_2/V_2}{I_1/V_1}$
Mass–Radius Ratio	
	$k = \frac{M_2/R_2}{M_1/R_1}$ $k = \frac{M_2/R_2}{6.73295 \times 10^{26}}$
Static Radius	
<i>Special Relativity</i>	$k = \frac{v^2}{c^2}$
<i>Isometric Form</i>	$k = \left(\frac{m_2}{m_1}\right)^{\frac{1}{3}}$
Time Dilation Forms	
<i>Normalized (vs. threshold)</i>	$k = \frac{D}{D_{\text{crit}}} = \frac{M/R}{6.73295 \times 10^{26}}$
<i>Raw (vs. other system)</i>	$k = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1}$
<i>GTD from normalized k</i>	$\frac{d\tau}{dt} = \sqrt{1 - k}$ (pace vs. flat spacetime)
<i>GTD from raw k</i>	$\sqrt{1 - k}$ (pace vs. other system)
<i>Recover D/D_{crit} from GTD</i>	$\frac{D}{D_{\text{crit}}} = 1 - \text{GTD}^2$
DeGerlia² Threshold	